ActInf Textbook Group ~ Cohort 6 ~ Session 7 (Chapter 5, Part 1) ~ 4/1/202

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 7 (Chapter 5, Part 1) ~ 4/1/2024 | 55:48

all right welcome back cohort 6 we're in
our first discussion on chapter 5
so let's jump over there um where does
anyone want to begin could be like any
page or figure or quote of five that
they remembered or that they're curious
about so first we'll just see any
thoughts or ideas people have about
five otherwise we can look to the
chapter and the prior
questions sure I I thought that um this
was a cool chapter because it for the
first time I think kind of gets a lot of
the message passing mapped on to um
specific biological systems that we
actually have quite a bit of prior
knowledge about um so I thought that was
pretty neat like to actually see the
sort of detailed example of the Pathways
in the basil ganglia and that kind of
thing I thought that was
great yeah it's go yeah I was taking a
look at that myself I don't know if we
could go through that a little bit
because I was just having some trouble
with the path tracing it might have just
been the model itself and the way that I
was misreading it
but uh uh figure 51 and
yeah 5 five is like the big overview
abstract it's gonna it covers the
prefrontal cortex graph the dopaminergic
the basil ganglia graph and the spinal
reflex arc graph and then earlier in the
chapter so now we'll go back to 51 and

so on earlier it's going to go through
those examples and then those examples
have a composition it
internally that's the graph and then
this is kind of the cool thing in the
interoperability is that because of that
compositionality internally you can kind
of do wiring across
graphs so that also speaks to the
compositionality um of the models and I
think maybe one theme that we'll try to
draw out and and explore and see
limitations of is like there's a
massive
underlying hypothesis of
computational Neuroscience which is like
computational
models can map biological
territories so there will be like some
regions or some
actual
tissues that either do that's more of
the realist angle or can be modeled as
doing that's more of the instrumentalist
angle they can be modeled as doing
certain
computations and if that's a viable
hypothesis then the maps that we make
with the math that we have are going to
be very apt if it's an inviable
hypothesis it could be extremely
misleading because there could be a
tissue and potentially the toolkit of
mathematical operators just isn't
adequate to make a good map of what that
tissue does leading to like a false
confidence about um the understanding of
of of what it's doing in the body or

like its role in development Evolution
all of that so so it's kind of like it
but that's so sublimated into the field
decades on that it's like of course
we're putting mathematical symbols on
top of pictures of
tissues but yet that's actually like
that's kind of the invisible elephant in
the room is like what what has happened
here applying these mathematical models
the graphical models and like using them
as overlays to make sense of the
functions of different
tissues yeah that's that's really
interesting I'd Wonder
um so perhaps kind of like in addition
to that one of the things that I had
heard a few neuroscientists raise is the
question of
whether the um whether the kind of known
connectivity maps in the
brain suit themselves to this kind of
explanation as well um oh I'm sorry
Andrew you have your hand raised would
you like to would you like to
go oh yeah uh thanks um yeah no it was
just a quick comment following from kind
of what Daniel is describing this sort
of like mapping the physiological to the
to the comput or mathematical I suppose
um I heard this nice
explanation uh it's in a previous uh
live stream with the Institute with uh
Ryan Smith the computational
psychiatrist and um it was just whenever
I was watching it it was like my first
time hearing like this kind of attempt
to directly relate neural activity to to

like I guess the mathematical equation
so he was like breaking down computation
of like variational free energy and
expected free energy and like equating
like basically attempting to equate like
whenever a term is subtracted from
another term that could be viewed as
like inhibition as opposed to excitation
with the like
addition and uh whenever something is
Multiplied in the equation we could view
that is like modulation in synaptic gain
so this just sort of I mean it it might
be taken a little too literally to just
state it as simply as that but it was
just yeah super fascinating to me to
think like oh like this is like really
attempting to map neural computation
with like you know a written symbolic
mathematical formula like subtraction
addition we have inhibition and
excitation and then we have you know
modulation with the multiplication so
anyway that that was it was so it was
more just a comment I can share the link
to that stream if anybody would like to
check it
out yeah like whether we're looking at
um gene expression levels of two loai
we're measuring with RNA seek or
something we're looking at the firing
rate of two different neural populations
or single neurons or like two
metabolites in the cell it's like okay
we're making measurements of some
biological properties and then that's
the kind of o to S mapping okay we have
the RNA seek data that's the observation

and then the hidden state is the true
gene expression level so that kind of is
the partially observable component so
let's just say that we're in a highly um
observable setting or whatever the case
may be we've gone past our measurement
into this underlying space of of how are
these two Loi related well one option is
like the mutual information between them
is zero like their co-variance profile
is just a scatter Cloud there's no
correlation so that point at least at a
first past there's not like an
interesting or non noise relationship
between the two variables not everything
is causally connected that's the
sparsity of the biological system so
it's like that's okay but then if there
is some kind of
informational
mapping then it's like okay at a first
pass does one go up when the other goes
up
or not you know is it a positive or
negative relationship is it a U-shaped
relationship there's not an infinite
number of
possible
mappings and then how complex does the
math need to be to capture that mapping
well that's kind of where we get into
this accuracy minus complexity question
is it enough to just say yeah linear
aggression or is it better to do it as a
um piece-wise defined function but like
well it's a linear relationship up to
this point and then it's
flat or would it be simpler to just fit

a single parameter linear regression and
then still just try to capture as much
variance as possible or should we go
with this two parameter model with a
slightly sharper slope and then a flat
Bridge me those are the model comparison
questions that make but the space of all
the possible
relationships does map pretty clearly to
these operators that we have
mathematically they
David yeah I was just wondering if you
can classify velosi as uh different
nodes in that
sense yeah exactly like a gene
regulatory Network the the nodes would
be the expression values of the Gene and
then the edges would be like a causal
connection different kinds of
edges
okay and and this also relates to the
the earlier point about the connectivity
maps and about different Notions and
different Maps that are made also this
is kind of like the genealogy of active
inference with the SPM package which is
exactly about well we don't only just
model the
anatomical structural connections that's
basically known I mean largely we know
where different tracts of neurons
project to um now there's also like the
electrical field propagation so on so
for for another time but um okay so we
have the structural connectivity that's
something you can just slice an image
and reconstruct in 3D but then what
happens is the dynamic causal modeling

approach says what are the causal the
statistical edges and then there's a few
different kinds of statistical edges you
might want to just take a first pass
which sections are correlated with each
other and then the edges basically
reflects a correlation value so it's
like these two have a 08 Edge between
them because they're 80% of the time
correlated um so that's one kind of
connectivity map another kind of
connectivity map would be looking for
the causal consequences of one region on
another so let's just say two regions
are 80% correlated but it turns out that
that's because they're both Downstream
of another region well in that case the
two Downstream
regions the 80% is kind of it's a
consequence of some other
[Music]
causal relationships so these are like
different kinds of maps that that um
connect nodes in ways that aren't
necessarily only the structural and
that's like a kind of tension that comes
up again and again where we have nodes
that are structurally connected to each
other like they're actually sending
neural
signals but for a given um pattern of
Interest that may not be causally
relevant and then on the other hand you
have nodes that are causally relevant to
each other so you want to model like a
causal Edge on the base graph but they
might not be directly touching through a
contact okay so then we it gets a little

um both like the simplest system slor
would be like all the anatomical
connections have causal efficacy and
none and there's nothing else so just
knowing the topology of the system as
it's actually
connected that is the computational
architecture however it's it's generally
not that way especially when making
Ultra reduced simpler models so like
here it's kind of a mixture of like on
one hand talking about the
projections of different anatomical
regions
but then on the other
hand that's not exactly or only the
edges that are described here like
there's other
projections that just aren't being
focused on
here okay I guess that relates to um
there was I'm trying to find the term in
the in the paper or in the book actually
um with the crossover
effect and I forget the name of what it
was what I just saw a connection with uh
interlan and crossover and there there
seems to be some correlation I just
don't know enough about the
uh the interaction if that might be
related to you know different nodes
interacting at different different time
periods as like things stretch out for
example in the frequency
parameters this is a great point this is
basically as presented here it's just a
model of like kind of the realtime fast
neural inference but we're not seeing

like the effect of glea or interlukin or
slower changes let let alone even just
synaptic plasticity so that's definitely
a big open area is how do you
incorporate this kind of runtime
inference
structure with like
development and and and neural and
immune development which change the
structure um another fun um paper I'll
I'll put it
here could neuroscientist understand a
microprocessor and this just you know
shots fired in the first
sentence why are people collecting all
this data I mean why are there massively
funded projects to sample synapses and
all this stuff well it's kind of related
to this belief that if we had more data
we better images better satellite images
of the territory then we would have
better Maps um so this
doesn't conclusively put that idea down
but they do something very fascinating
and empirical which is we know the
topology of a
microprocessor and there's simpler
microprocessors that can be simulated in
silico that's kind of meta but we have
the full processor simulation being
emulated at like a tremendous overhead
right on a more modern processor and
then they're able to take neural
recordings from the microprocessor and
you could look at like the activity of a
transistor you could also look at like
the local field potential around a
region by by averaging out the the

electrical
activity and then you could do single
and double lesion experiments and that
kind of like loss of function is classic
like in genetics like that's what people
do for a long time it's like gain of
functions and losses of function and
looking at single and pairwise losses of
function and then they presented that
data
set to neuroscientists and used their
kind of general information processing
algorithms and we see this kind of
coming up like in integrated information
Theory like they're talking about
microprocessors and they're talking
about neurons because both of those are
being abstracted to the graphical
Network information Dynamic setting and
um it's not to say someone couldn't
analyze this information usefully it
just turns out that there are like
things that are causally connected
because we know the software running not
to say that they software running on
neurons but again it's an allery there's
things where we know what computation is
happening and the lesions are
uninformative about it and there's it's
just there's there's false positives and
there's false negatives that's not to
say it's uninformative experiment
but it it it really calls into
question a simple or kind of uncritical
mapping between how things are
structurally connected in the real world
anatomically and then what's the causal
architecture of the system this comes up

in the ACT INF side like the whole marov
blanket discussion our marov blankets
reflecting the spatial and the temporal
boundaries of
things if so where Andor do marov
blankets just more generically reflect
boundaries in state space which might
coincide with a spatial and temporal
boundary but it doesn't have to there'd
be boundaries in state space that would
be um within what would be called a
spatial or temporal boundary and and
vice versa it would all depend on your
your sensitivity thresholds and whether
you were trying to capture 80% of the
variant or 99 or
99.999
so that's what's so interesting again in
this chapter 5 which is like we had the
intro chapter one low road High Road two
and three chapter four with all the math
and getting to the actual kernal of the
generative model and then now that is
kind of perfect in and of itself but it
hasn't really been applied to any system
and then chapter 5 immediately is kind
of like here's some of the amazing
insights you get
by applying active inference to real
territories and here's some also of the
fundamental challenges and
limitations that that come into play
when we are talking about maps and
territories rather than just kind of the
pure science of map
making because you could just make these
kind of pure
um maps of just just well I'm imagining

a territory that's you know
there's a street map on the head of a
pin but then you can't necessarily build
that or measure it in the real world but
you could talk about it in chapter four
um yeah I guess that kind of speaks
towards like some of the granularity as
well in terms of like in silico versus
in situ if you're doing a comparison you
know going back to the transistor model
you know how fine grain can you get I
was just wondering if there's any
correlation uh in that sense as well so
like you know the smaller the
transistors that get you know nanometer
wise um is there any correlation there
with you know Inu with um with the
Dynamics of of neuron
interaction yeah I think a great example
of that empirically is like it's often
this is like early 1900s the debate was
like is the brain composed of cells that
have gaps between them or the reticular
theory that was the neuron
Doctrine there are cells that are
separated and then there's the reticular
Theory which is kind of like maybe it's
all like a
mycelial
connected um W with actual direct
connections um and then that's the GGI
stain and ramonica Hall and they were
like no you can stain a cell and it
fills out the cell and ends there so
that was that was like a big W for
neuron Doctrine then you know there's a
little bit of complications there's some
big holes and St and there's Parts where

they are connected and so on but but
largely the cells are separated okay so
then the neuron became understood as
kind of like the unit of
modeling but then there's several issues
there um one is that's a huge model even
for a simple brain region so often we're
modeling populations of neurons like
these nodes represent not single neurons
but populations of neurons and that's
the concept of like the population
encoding and those are easier to work
with in terms of coar graining
because you're course grading to fewer
regions and also the statistics of
individual neurons are very spiky and
noisy whereas when you're looking at the
population statistics of a pool then you
can model it much more easily with like
continuous distributions and like rate
encoding it's more like looking at an
analog signal that's smoothly turning up
and down rather than like spiking but
then going down a level people have
found just incredible computational
Dynamics in dendritic
Arbors so even just choosing the neuron
as the unit of analysis like you can
look at the computations or whatever you
want to think about it as at a sub
level like if a neuron had two branching
sets of dendrites you could look at the
summation Dynamics and the Decay
Dynamics at a really fine scale
level and then at that point like the
neuron would be kind of your higher
level
outcome and that's part of the

multiscale um processes that are
happening again with a map territory
none of this is an issue it's like well
if we wanted to do that um millimeter
scale analysis of the sidewalk we could
or we could just have the 1 kilometer
map but then that's where um realist
approaches to computational Neuroscience
are are
in a in a difficult
position because it's so clear even at
the single neuron level that you're
corar graining and abstracting over like
many levels of
function right I mean I I see that and
like I mean I haven't done too too much
research but just from what I've read um
you know in terms of like zor Gates and
how an individual neuron interacts once
you get a field of neurons as an array
you know how can you tell what gates are
opening closing and what's interacting
or cascading across the entire
field yeah like right now the nodes are
AR just in this image the nodes are
arranged to resemble the histological
layers of the cortex the melan preval
cortex um but all the causal
relationships are topological so like
technically you could like kind of
rearrange the nodes on the page as long
as you kept the connectivity and that
wouldn't change the computational
function here it would just make it look
less like the layers of the cortex would
you oh yeah good would you have any
gradient like layer shifting so would
you have you know any of the modalities

shift between the layers because of that
interaction or would it be like would it
remain in situ and just the connections
themselves would be the things that are
changing that's a good question I mean
if your map were purely
topological then geometric changes
wouldn't matter however if you did want
to include like bulk propagation of
electrical flows or spatial effects then
that would be a different kind of
model so then that which is not even
represented here and then there's like
different ways you could incorporate
that into the model like you could give
each each node could have a position in
in a coordinate system and then you
could have another component of the
model that had like a propagation
through that coordinate
system and then this then this would be
have why it's so important to have the
portfolio of models you say okay I have
the I have the 100 parameter model here
that's purely topological and then I
have the 109 parameter model that has
this and then the 120 that has this
other thing and then accuracy minus
complexity um Bas information theorem we
want to reward explaining the variance
in the data we want to penalize having
more
parameters how much should we penalize
having more parameters well that depends
on the modeler's sensitivity are they
seeking to explain a higher and higher
fraction of the variance at the cost of
a model increasing parameters or are

they looking to actually take just a lwh
hanging fruit and have a kind of sparer
model accepting that it will explain
less variance there's no a prior answer
because there's a setting where you want
99% variance explained at all costs and
then there's a setting where you want
four variables and no more and whatever
variance you explain is just what you're
going to get with four so those are just
the the pragmatic considerations of of
making a given
model great thanks
okay so here we're in the um cortical
layers um as always the disclaimer this
is just the Maman nervous system plan
analogous and other functions are
carried out by other like diverse
nervous systems so I think it'd be epic
to have like a chapter 5 insect
supplement especially because it's a
fully observable neural system you just
lay it flat and just take the picture so
I think there could be Mega insect brain
modeling but here's the cross-section of
the neural tissue and here are those six
layers here's the surface of the
prefrontal cortex and then here's the
columnar
architecture which are these kind of
hexagonally tiled
columns that are the basis of
of for example thousand brains by Jeff
Hawkins this
um this is an interesting work
definitely I wonder about what they know
about actim um but it has to do with um
the the parallel parallelized

decentralized architectures and kind of
the um
columnar um architecture
okay Peter I'm going to look at
this
awesome
yeah wow I mean you know these
are the silent elephant in the room
isn't that great
yeah it's like can we really just take
highlighters and like shade different
parts of real tissues well this one's
doing
prediction okay but can can in a
different cognitive setting or a
different like attentional
regime is it always doing predictions
it's like about what so those and uh you
know where are the glea there's all
those slower questions so that's that's
always really fun but this
is this is great and it's looking at the
um relationships within and across the
columns
yeah one thing I really like about this
paper um and both these papers actually
is that for some of the underlying like
architectural stuff like looking at the
connections between laminer gradients
across the brain and that kind of thing
um it depends on the track tracing work
of Helen barbas um whose work is like
some of the most like detailed and
exacting that I've seen in the neural
literature um so like if we're going to
try to do this kind of mapping onto
biological systems uh which seems
difficult enough you know with all all

the things that you you said Daniel it's
at least nice to
like have uh have a sort of tracing of
the actual circuits that it feels like
we can kind of take to the bank you
know yeah like we we take the tracing to
the bank as kind of a fact and then that
is like our that's like a very strong
very informative prior for causal
relationships like all things considered
where there's an actual
Edge it's pretty viable to suggest that
there's a causal efficacy to that edge
somehow and then when there's no causal
Edge there could be you know telepathy
there could be like long range bulk
electrical transport there could be
relationships mediated through other
local connections but I mean on the
first pass nodes that aren't connected
it makes sense to exclude those as
having causal relevance for each other
at of first pass so the structural is a
great or if we were talking about a
communic a network it's like we know
which people talk to whom or which
computers talk to whom that's a starting
position for the causal architecture of
the
model so that's like that is I think an
important role for the neuro anatomy in
this um okay so
here it's going to be focusing
on one columnar
slice and it's going to show it three
different ways moving from on the left
we have the labeled cell types so this
is the most

neuroanatomical where the edges are
reflecting the actual projections
anatomically again not all of them Etc
but this is just describing Anatomy it's
not even a base graph yet then in the
middle we see kind of a classic
predictive
coding architecture
where we have m's for means and then
like e for errors and then we see the
eyes for a given level and i+ one for a
higher level so and and the X and the V
subscripts so this notation which of
course is resembling but it doesn't
pretend to exactly recapitulate this is
the predictive coding type computational
map on this territory which is is you
have top down predictions and then you
have incoming bottom up sensory evidence
for
example uh which could be coming from a
thalmus or or lower regions and then you
have an error
term that's just checking the
differencing so this is going to be a
lot like what we see in the spinal
reflex arc but this is happening in kind
of like a more cognitive setting the
more general setting in predictive
coding and then here on the right
we get to the
PDP this is the most kind of chapter
7ish generative model and although there
are some letters that are a little bit
outside of the figure 4.3 broadly we're
talking about the hidden State
inference so we kind of move from the
most anatomical to the very highlevel

computational predictive processing
Paradigm active inference is basically
predictive processing with action and
then but this is in a continuous setting
and then it's brought into the PDP
setting So within a column whether we
think about it like kind of purely
anatomically whether we think of the
column as doing predictive coding
processing or whether we think of it as
doing a PDP with like it's carrying an S
estimate through time and then
observations are coming in and here
there's kind of anticipation the O Tilda
through time and expected free energy
and all that
PDP if we think about that within a
column there's an interesting way to
talk about the relationships among
columns which is that they're
arranged side by
side in the
cortex and each one of the columns is
like one layer of a
hierarchical basian model and then you
have lateral connections which that was
a a great paper um Peter that
showed these kinds of lateral
connections go between the layers and
then also just like mentioned like where
we see summation like excitation
empirically we can associate that with
addition where we see inhibition we
could talk about that as subtraction of
like a a likelihood or like a rate
constant and then when we see
multiplication which is kind of the same
as division just whether you do like one

divided by multiplication division is
like a neurom
modulation because it doesn't directly
summate or subtract it's like a a
coefficient that multiplies that so
that's neurom modulation and then here
is like we have like a three level
nested model so the six layers perform
the role computationally of one layer of
a nested basian model so this is like
we're we're predicting a three-digit
number and we're breaking down that
problem factorizing it into the ones
tens and hundreds place and then here's
the the top down prediction is you know
what it is and then the bottom up is
coming up and that's how the these three
layers even though each one is
only holding one digit that's how the
three of them you know just
super broadly would would um do a three-
layer model
here okay so now we're going to
trace this
path this is the the descending motor
prediction coming out of a for example
motor qual region it's going to be
projecting down to um regions involved
in motor control and neurons that have
motor right in their name so they have
to be doing it
right here's the projection
down now we're going to go from um Mew
till
here
basically the bait cells might be
interesting to look at them um sending
down the

paramal
tract and it is synops onto
this essentially combining and differenc
her here we have observations why coming
in from
proprioceptive tissues in like the elbow
so there's like all these different
kinds of touch and appropriate receptive
receptors like some of them look like
onions so when they're depressed they
activate other they have different like
shapes it's kind of cool that is coming
in and then it is being passed along
from the dorsal horn through an internal
synopse within the spinal cord to the
vental horn
population and then there's a differ
thing and this is just a first pass
approximation but where the descending
prediction is ex exactly matching the
incoming propri reception no action is
required and um that that kind of makes
sense that's consistent with this
differential driven approach to
basically making
moves um and
then if you
have I guess it's in a different chapter
where it shows the um the set point
changing and it shows how the um how the
the the motor Dynamics are but this is
just this is like a kind of sub motif of
predictive processing but this is just a
frame differenc sir and and this is also
very related to um PID control you set
point and
then you know again two a first
approximation if the temperature of the

room is if you set it to 72 and it's at
too why would you take an action maybe
you have a more sophisticated model you
have a super strong belief it's about to
get hot and you know it takes time to
turn on the air conditioner so even
though it looks perfect right now you
actually want to turn on the air
conditioner so that it counteracts like
alost statically a future prediction but
just at a first pass if you don't have
that kind of allostatic
anticipation if something is at the
level where it's set to be no further
action is taken David
yeah I mean you you had mentioned also
in terms of like message passing like
what would the dopam energic response be
would it be an initial kind of like
packet you know that would be sent that
would be a priority of you know the
cognitive response itself before the
decision is made and how might that
actually you know be false signaling in
some some cases if you're unaware of you
know in a priority situation for
example
let's this is
one a wonderful title and also very
interesting paper and and I think
there's some other ones on this
um
topic but Columbo investigates from like
a history and philosophy of science
perspective on the on the dopamine
discourse the metad dopamine studies
about well
dopamine I mean if we look it up let's

predict is it going to call dopamine a
reward
molecule pleasure satisfaction
motivation what is dopamine
does desire so it's framed in this kind
of salience motivation
reward reinforcement but but there's
other functions for it and in the body
like we were talking about before um
recording like with DOP mean and in the
immune system and all this but Columbo
kind of looks at and he says look the
empirical evidence even for the
dopaminergic brain regions like we're
about to get to in the basil ganglia
like there's populations of neurons that
do have a correlative firing pattern
with reward there's also patterns that
are firing more with a surprise model so
there's one other Columbo paper that's
like an unrelenting
pluralism it's just it's like it's just
very interesting like and I see as as a
a way that um forward-looking
philosophers of biology can actually sty
me I don't know effectively or not but
um like kind of absolutism or like
closing the book on what dopamine does
when it's like actually we already have
enough evidence to say that there's a
plurality of functions for
dopamine so then further empirical
evidence can can be brought into a
plural framework where dopamine plays
different computational or cognitive
roles rather than the debate needing to
be like what computational role does
dopamine Play We Found evidence that it

plays a role in prediction dopamine
plays a role in
prediction but if we have already kind
of laid the top soil with
pluralism then it's kind of like all
right great more points for pluralism um
more points for this function of
dopamine but but it can't be resolved in
in principle what dopamine does because
we already know that we have some
tallies across different functional
categories um there's no dope mean in
this um circuit yet no no Dove me
directly okay I don't think I don't know
exactly when neurotransmitters are here
but um dopamine plays more of a
neuromodulatory role than um especially
in the it plays a role in the central
nervous system that way but in the in
the peripheral um system and then like
famously at the um neuromuscular
Junction so this synapse here is like
where you have acetal
choline okay so this is just kind of a
simple
um differencing algorithm but
differencing algorithms are
like I mean they're they're they're the
heart of many things happening and also
of course going to to the history I mean
predictive coding predictive processing
predictive compression was invented by
frame differencing video compression in
like the 197s and 80s and
then there is the r and
Ballard Mega
citation 1999
paper that was like oh classical means

whatever was done before right so then
there was these extra classical
effects in the visual cortex and then
they were like wait a minute if we think
of what the visual cortex of doing as
protction then that explains why there
are these extra classical phenomen
that are not being evoked by the stimuli
because they're being invoked by the
predictions so then that kind of like
broke I'm sure there was like a little
bit before too but this big citation
kind of broke the
wall with just the idea that like hey
even primary sensory processing is
predictive so it's only been 25
years and that's the interesting thing
is moving from the concept of well
signal processing and sensory
transduction is related to processing
and Distilling and extracting the
information from the real good sensory
data it's so much out there and we're
just getting the good stuff from that
versus actually the sensory data coming
in are like lagged noisy biased partial
Etc however they just kind of help
tune an ongoing multimodal generative
mod model and that's why our visual
field has color everywhere and no blind
spot attention all this other stuff
because we're not just like processing
and sharpening the TV signal and then
displaying it on the inner screen
there's an inner generativity that's
multiple hierarchical nested levels away
from the primary sensory
information but this is a this may have

made sense to look at before this one
because this is a simpler Motif it's
kind of a core Motif is this just
concept of like take the difference and
then just use the sign of the difference
positive or
negative like at a first pass that's the
direction to go I mean if there's a
speed limit and you're over it go down
you're under it go up unless you have
like some other higher order reason to
do other than what seems
obvious which is right what which is
very like cybernetic right yeah it's
it's just like for it's like bread and
butter first order
cybernetics and then second order and
higher order cybernetics would would
kind of be like where and when and how
does the first order
cybernetics lead to like a death
spiral um and then also could relate
maybe to system one and system two and
just on a first order take theistic but
then to to override heris which is now
this is exactly where it takes us here
is a more
challenging um path so here it's an
interesting like visualization I think
of this one is kind of like the scales
of Justice in a way because it's not
saying that the left and the right basil
ganglia have different
topology it's actually just like showing
to modes of
function that are like it's like a RAR
Shack blot unfolded and then the two
sides are like the two different modes

but it's not it's not saying that it has
these two different architectures on
either side but basically there's two
path they could the indirect and the
direct pathway are both bilateral as far
as I understand but they're just
separating them out here so these are
different brain regions um with their
acronyms it's a simplification
okay
um in the indirect
pathway observations are coming in um
and then there's o Tilda o through
time of a given policy and here's e The
Habit this is the prior on policy and
then basically those
indirect the the prior on policy habit
is just being passed on to what we
choose
so this is just kind of carrying forward
our prior on what we do and then either
choosing the best ranked option or
sampling from that policy posterior so
it's kind of like saying let's not do a
policy update let's just make our policy
posterior or policy prior so no F no
calculation of expected free energy no G
coming into
play um equivalent to not paying
attention to observations to incoming
observations or expected future
observations so it's Mega simple and
there might be an Adaptive habit that in
a given environment for which that habit
was developed adaptively for that's
super
simple on the other hand there's a
direct pathway mode that they're

analogizing as actually doing the
expected free energy calculation G and
updating the policy prior into the
policy posterior according to equation
2.6 the whole part about pragmatic and
epistemic value so it brings in a ton
more variables and
computations but it
sharpens the prior into the posterior
based upon the incoming observations and
expected future observations so that's
kind of like the deliberative system to
thinking whereas this might be
analogized to
theistic faster system one thinking
David yeah how would that affect like if
you're skipping that first part would
that direct to the map of like the
ontological directly without then prior
updating so you're kind of going without
a map in that sense on The Logical side
of things
right that's interesting like habit is
kind of like map free it's kind of like
no just this is my turning distribution
I I I go 50/50 left and right and I'm
just sampling from that wherever I am
versus like the more deliberative would
be like well if I turned left then this
is where I'd be if I turned right this
is where I'd be so the habitual is a
more implicit
ontology it inherits from the the
ontology that that supports that habit
whereas the
explicit all basically
enumerates the onology and creates like
a

local projections that could be based
could could be also leveraging
information like from external Maps or
other Maps
Andrew oh yeah I was just um wanting to
relate it to like the the authors in the
textbook relate um like the the comp
computation and use of the empirical
prior e in policy selection as like
being akin to like a model free system
versus um the use of G and computation
of that to inform like um policy
selection and policy comparison is like
modelbased system which kind of made a
little bit more like helped me track
that a little bit more as someone with
more of a data science rather than
Neuroscience background but um yeah no
it's it's super interesting I'm trying
to remember another paper that I had
read that you know with with with the
ACT in terms of like confidence and so
on there there was one paper I'll try to
remember it but it Likens like an agent
in its environment who has confidence in
its model and also receives some kind of
confirmation that its model is actually
helping it to realize its preferences
it's as if it becomes more confident and
its um ability to use a model to compute
G parentheses
in order to realize its preferences
right versus if it repeatedly hits like
obstacles there might be a sort of
regression into like oh my my confidence
in my model is is very poor it relates
to that um relates back to that gamma
term is a kind of like Precision or or

or or
temperature uh term that gets involved
with the G computation in any case um
yeah whenever the the agent becomes less
um confident in its model and its
ability to actually compute uh like like
useful policies it starts to kind of
regress back into like habitual
behaviors that it is previously relied
upon and then from there I presume like
you know over time you might very well
end up modulating your your pro your
your um have your habits as well like if
we view those as like in values over
over policies that are sort of ready to
go like those get changed too over time
but uh anyway I'm kind of rambling at
this point but those are great
points this has always been where where
I get hung up on with a model
fre which is model fre just means
simpler
model so you know when something's like
next Generation this or or this you know
none of this but then it's like it just
it's hard to keep track because people
are using using those as kind of like
relative
terminology but it's even used in
chapter
10 I think it would make sense to to
have mentioned model 3 in chapter 5
because it literally is mapping to this
arbitration and then just in the last
minutes I mean
5.1 describes work in the act in space
and by others with linking the
computational function oh and dopamine

is playing the the um judge jury
executioner role with a dopaminergic
tone setting the difference between
habitual and deliberative behavior but
there's a lot of like there's a lot of
effects of dopamine so that's why it's
not so simple as just taking like you
know you know yay dopamine or
nay but these kinds of things some some
with friston and others others just more
generally
um the pupil I think the icade and the
pupil modeling is really interesting
those are things that most people can't
control consciously and they're not
aware of consciously but they reveal a
ton about the generative
model
um and
then they kind of uh allude to the the
the um the uh favorite synthesis topic
of this textbook which is continuous and
discreet hierarchies coming together
like every chapter it it kind of comes
up and then here we have
it if we use that the rightmost form
here the PDP discret time like form so
we have a discret neuros symbolic model
here and then down at the Frontline
workers we have the continuous sensory
motor form and then we have the the
policy
selection
module is not directly linking down to
the spinal cord so it's not like
prefrontal basil ganglia and then sends
the message down to the spinal cord it's
like the

prefrontal and the basil ganglia are in
a dialectic and then there's a
descending motor prediction that
reflects multiple scales of the
resolution of that
Dynamic by just passing down the set
point and then that is enacted
autonomously with What's called the
reflex
arc they bring up a ton of other
questions
cool topics for future
discussion
okay thank you everyone see you later or
next time have a good one thanks
everyone